

Polysaccharides

Found in all plants, polysaccharides are multiple units of sugar molecules linked together. From an herbal point of view, the most important polysaccharides are the “sticky” mucilages and gums, which are commonly found in roots, bark, leaves, and seeds. Both mucilage and gum soak up large quantities of water, producing a sticky, jelly-like mass that can be used to soothe and protect irritated tissue, for example, dry irritated skin and sore or inflamed mucous membranes in the gut. Mucilaginous herbs, such as slippery elm (*Ulmus rubra*, p. 145) and linseed or flaxseed (*Linum usitatissimum*, p. 109), are best prepared by soaking (macerating) in plenty of cold water. Some polysaccharides stimulate the immune system, for example acemannan, which is found in the leaves of aloe vera (*Aloe vera*, p. 60).



Slippery Elm
(*Ulmus rubra*)

Glucosilicates

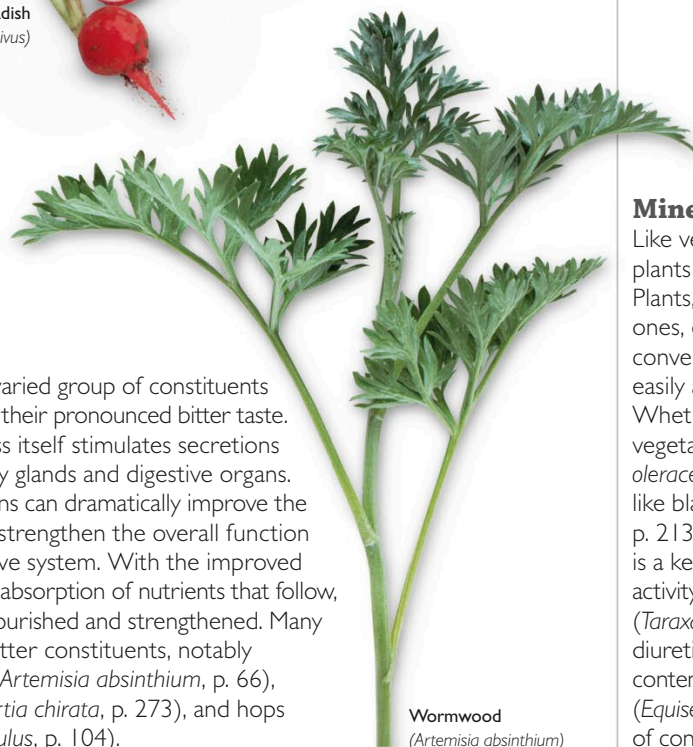
Found exclusively in species of the mustard and cabbage family, glucosilicates have an irritant effect on the skin, causing inflammation and blistering. Applied as poultices to painful or aching joints, they increase blood flow to the affected area, helping to remove the buildup of waste products (a contributory factor in many joint problems). On eating, glucosilicates are broken down and produce a strong, pungent taste. Radish (*Raphanus sativus*, p. 261) and watercress (*Nasturtium officinale*, p. 239) are typical glucosilicate-containing plants.



Radish
(*Raphanus sativus*)

Bitters

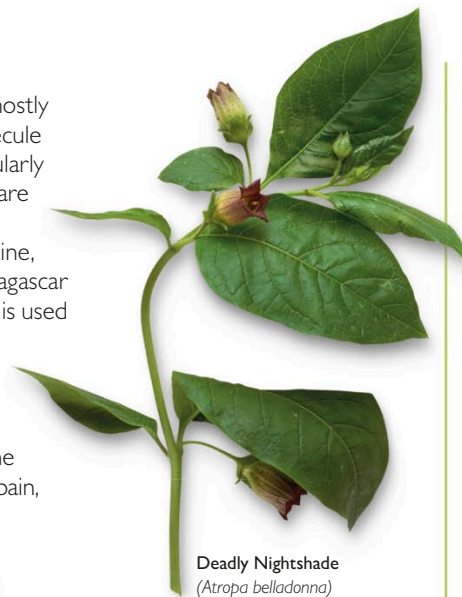
Bitters are a varied group of constituents linked only by their pronounced bitter taste. The bitterness itself stimulates secretions by the salivary glands and digestive organs. Such secretions can dramatically improve the appetite and strengthen the overall function of the digestive system. With the improved digestion and absorption of nutrients that follow, the body is nourished and strengthened. Many herbs have bitter constituents, notably wormwood (*Artemisia absinthium*, p. 66), chiretta (*Swertia chirata*, p. 273), and hops (*Humulus lupulus*, p. 104).



Wormwood
(*Artemisia absinthium*)

Alkaloids

A very mixed group, alkaloids mostly contain a nitrogen-bearing molecule ($-NH_2$) that makes them particularly pharmacologically active. Some are well-known drugs and have a recognized medical use. Vincristine, for example, derived from Madagascar periwinkle (*Vinca rosea*, p. 282), is used to treat some types of cancer. Other alkaloids, such as atropine, found in deadly nightshade (*Atropa belladonna*, p. 69), have a direct effect on the body, reducing spasms, relieving pain, and drying up bodily secretions.



Deadly Nightshade
(*Atropa belladonna*)

Vitamins

Though often overlooked, many medicinal plants contain useful levels of vitamins. Some are well known for their vitamin content, for example dog rose (*Rosa canina*, p. 263) has high levels of vitamin C, and carrot (*Daucus carota*, p. 201) is rich in beta-carotene (pro-vitamin A), but many are less well recognized.

Watercress (*Nasturtium officinale*, p. 239), for example, contains appreciable levels of vitamins B₁, B₂, C, and E as well as beta-carotene, while sea buckthorn (*Hippophae rhamnoides*, p. 220) can be regarded as a vitamin and mineral supplement in its own right.



Dog Rose
(*Rosa canina*)

Minerals

Like vegetable foods, many medicinal plants provide high levels of minerals. Plants, especially organically grown ones, draw minerals from the soil and convert them into a form that is more easily absorbed and used by the body. Whether plants are eaten as a vegetable, like cabbage (*Brassica oleracea*, p. 180), or taken as a medicine, like bladderwrack (*Fucus vesiculosus*, p. 213), in many cases the mineral content is a key factor in the plant's therapeutic activity within the body. Dandelion leaf (*Taraxacum officinale*, p. 141) is a potent diuretic, balanced by its high potassium content, while the high silica content of horsetail (*Equisetum arvense*, p. 205) supports the repair of connective tissue, making it useful in arthritis.



Dandelion
(*Taraxacum officinale*)